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Dear Editors,

We are pleased to submit our manuscript entitled “Experimental Investigation of Flow Boiling Heat Transfer in an Annular Minichannel with ZnO-, ZnO/PMHS-, and Al₂O₃-Modified Heated Surfaces” for consideration for publication in *Energies*.

The manuscript presents an experimental investigation of subcooled flow boiling of distilled water in a vertical annular minichannel equipped with interchangeable smooth and surface-modified heated tubes. Copper and stainless-steel substrates were tested as smooth references and after modification with ZnO or ZnO/PMHS, while an Al₂O₃ coating was additionally investigated on stainless steel. The study combines surface characterisation, including contact-angle, roughness and SEM measurements, with a cylindrical data-reduction model used to determine the local wall temperature and effective heat transfer coefficient. The thermal performance of each modified surface was evaluated through a pointwise differential comparison with the corresponding smooth reference at matched heater-power settings, flow conditions, power branches and axial positions. The results demonstrate that ZnO on stainless steel was the only tested modification that produced a consistent beneficial response at both investigated mass flow rates. It simultaneously increased the local heat transfer coefficient and reduced the wall temperature. The remaining coating–substrate combinations did not provide robust enhancement. The study therefore shows that the effect of surface modification depends on the combined interaction between coating morphology, wettability, substrate material and operating conditions, rather than on the nominal coating type alone.

We believe that the manuscript fits the scope of *Energies* because it concerns experimental heat-transfer enhancement, multiphase flow in compact thermal systems, surface engineering and the development of advanced cooling technologies. The results also provide a reference data set for future investigations involving refrigerants, dielectric liquids and other cooling fluids.

We confirm that neither the manuscript nor any part of its content has been published previously or is currently under consideration for publication elsewhere. All authors have read and approved the manuscript and agree to its submission to *Energies*.

Thank you for considering our manuscript.

Yours sincerely,

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